

PROJECT PROPOSAL

Predicting Heart Disease from Clinical Measurements

1. Problem Statement

Heart disease is a leading cause of death, and early detection can save lives. This project builds a machine learning model that predicts whether a patient has heart disease from routine clinical measurements such as age, chest pain type, blood pressure, cholesterol and exercise test results. This is a binary classification problem.

2. Dataset

We use the Heart Disease UCI dataset (UCI Machine Learning Repository), which combines patient records from four hospitals: Cleveland, Hungary, Switzerland and VA Long Beach. It contains a mix of numeric features (age, blood pressure, cholesterol, maximum heart rate, ST depression) and categorical features (sex, chest pain type, resting ECG, exercise angina, slope, thal). The data is publicly available for research and is anonymised.

Data source	UCI Heart Disease dataset (4 hospitals combined)
Rows × columns	920 × 16
Target	num → binary target (0 = no disease, 1 = disease)
Task / metric	Binary classification; F1-score and ROC-AUC

3. Target Variable

The original target num ranges from 0 (no disease) to 4 (severe disease). We convert it to a binary target: 0 = no disease and 1 = disease present (num > 0). The classes are fairly balanced (about 55% disease and 45% no disease). We use F1-score and ROC-AUC rather than accuracy alone, because failing to detect a patient who has the disease is more costly than a false alarm.

4. Data Challenges and Planned Handling

- **Missing values:** thal (about 53%) and slope (about 34%) have many missing values and will be imputed; ca (66% missing) will be dropped.
- **Hidden missing values:** zeros in cholesterol and blood pressure are medically impossible and will be treated as missing.
- **Outliers:** kept, because the values are medically possible, and handled with RobustScaler.
- **Feature work:** a new feature hr_pct (share of age-predicted maximum heart rate reached); the hospital column is dropped so the model does not learn the hospital instead of the disease.
- **Leakage:** all preprocessing is built with scikit-learn Pipeline and ColumnTransformer and fitted on the training set only.